

Urban Design For Community Queen's Lawn & Dangoor Plaza

Bdobe Outdesign

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Context

Inspiration

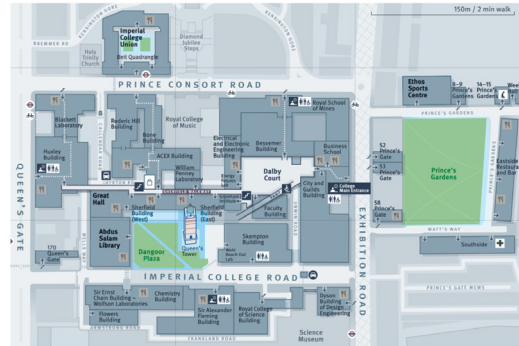
From History

Limited Number of seats

Personal Experience

Limited Number of seats

History -- Build and Rebuild



- The queen's lawn is the **only greenery** on campus, and it is the.
- **Circulation** is high during lunch time.
- **Good location** near library and surrounded by several teaching buildings which is **center** of the imperial south kensington campus

Approximate Area

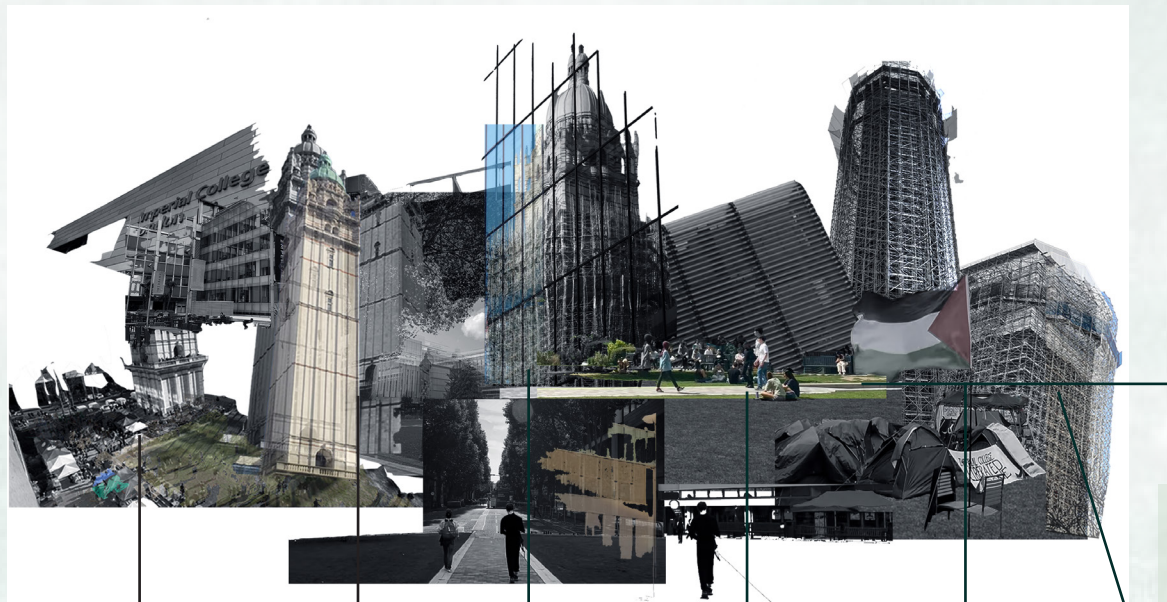
1700 m²

Population

3,1000

Number of Benches

20



1887 – Imperial Institute **Established**
• The Imperial Institute is founded to commemorate Queen Victoria's Golden Jubilee.

1957–1963 – **Demolition** of the Imperial Institute
• The majority of the Imperial Institute buildings are demolished to make way for the expansion of Imperial College London.

1970s–1990s – Landscaping and **New Campus Layout**
• The surrounding area is landscaped and converted into the Queen's Lawn, creating a green space for students and staff.
• The Queen's Tower is preserved as a standalone structure, becoming a key historical landmark of the campus.

2000s–2010s – **Expansion** of Campus Activities
• The Queen's Lawn becomes a popular site for college events, including student fairs, formal balls, and weekly farmers' markets.

2021 – Construction of **Diagonal Path** on Queen's Lawn
• A new diagonal pathway is added across the Queen's Lawn, improving campus connectivity and accessibility.

2017 – Restoration of the Queen's Tower
• Conservation efforts take place to restore and maintain the Queen's Tower, ensuring its structural integrity and historical significance.

5W & 1H

Who

Students and Staffs

When

Eating, Talking, Chatting

How

- Certainly not much use of the grassland
- Students need better place for group chatting, eating
- Increase interactions between people

What

- Transformable seats and tables with different functionalities
- Relate to imperial culture

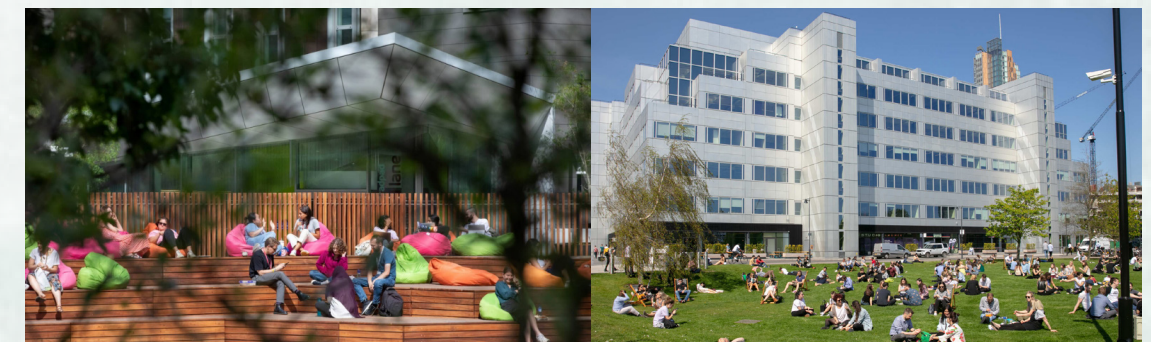
Where

Dangoor Plaza & Queen's Lawn

Why

- Motivate interactions between students and stuffs
- Solve the problem of limited seats and tables
- Provide place for leisure activities

Research of Existing Examples



Imperial College London- White City



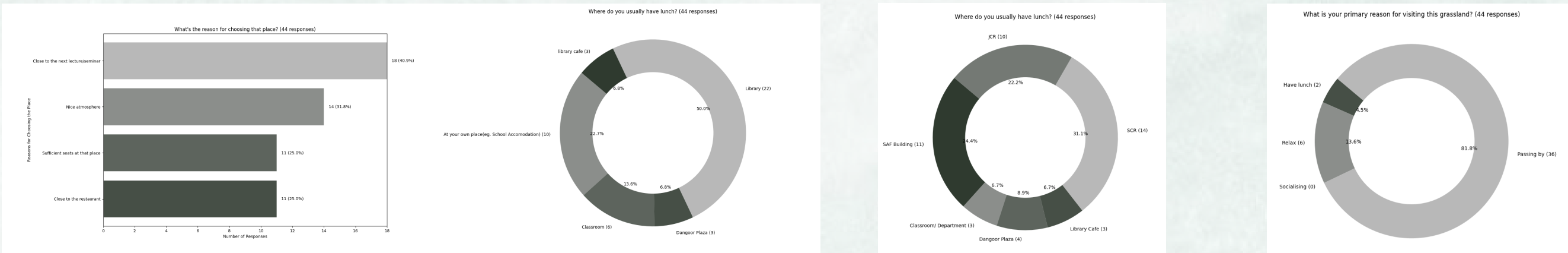
Urban Movement Design Debutes- Rome

To provide more space for study, group meeting and leisure activities with better atmosphere!

Primary Research

Survey Results

By removing any bias and assumptions, we designed a **questionnaire** about the current environment and **potential problems and changes**. The surveyed samples are mostly Undergraduate students from Imperial and a few Master degree students, which in total has a number of 45 students. The data collected is crucial for us to analyse the findings of the questions we asked, and potential insights and suggestions.



Most of the users choose the place because it is **close to the next lecture or seminar**. **31.8%** of the users choose due to the **nice atmosphere** and the rest are either because of the **sufficient seating** or their location is close to the place.

Half of the users choose to go to library for **self-study**, as well as **22.7%** decide to study at their **own place**. The minorities are more willing to go to classroom and **fewest** users are likely to go to library café and Dangoor Plaza.

Most of the users like to go to **SAF building, SCR and JCR** for lunch (24.4%, 31.1%, and 22.2% respectively), whereas the rest like to go to their classroom, library café and Dangoor Plaza.

Over **80%** of the interviewees visit the grassland just because they have to **pass through** it to reach destination, followed by **13.6% of them enjoy** relaxing on the grassland. A few people are willing to have lunch there (**4.5%**).

Research Overview

Aim: Observe the daily behaviour of people while **crossing or passing** by the Dangoor Plaza. Think and ask questions to collect sincere and practical **answers and insights**. Identify a targeted user with its perspective concerns about the place and gain a better understanding of the requirements and demands of the customers by analysing their **experience and behaviour** and performing co-design activities. Finally generate design ideas to satisfy user demand.

Context



Observation

We visited Dangoor Plaza **3 times every week** noticed that it is **overcrowded at Tuesday** during Farmers Market. A **lack of seating** was observed and the area is not very welcoming when there is no sun.

Interview

10 undergraduate students from Imperial aged 18-19, **3 maintenance staffs and a professor** were interviewed. During the interview, we discussed current and potential problems of the lawn. The staffs offered ideas about the **history** and constraints of the plaza, where the professor provided ideas of future thoughts and designs.

User Experience

We designed **4 personas** according to different roles at the school, including **1 extreme user and 1 extreme non-user**, while the rest focus on different aspects of their needs. **Mapping** their needs and concerns help us with **insights** to focus on further ideas of the design process. Leading us to think about specific needs, places and facilities.

Ideation



Observation

Insufficient Seatings

Tuesday Noon, 13:00-14:00, during **farmers Market**.

People struggled to find seats for eating, they **sit on chairs beside AF building** and the benches around the dangor plaza and eat with no tables. Some people use the **sculpture as the table** or **sit on the stairs** outside Chemistry Building.



Bad Condition

The condition of the grassland is **poor**, where **little grass** grows well. Some of the tables are placed in **muddy conditions**, affecting user experience. All the existing tables are placed **without any waterproof facilities**, leading to high potential risks by weather.

Communion can be **difficult in rush hour** (during lunch and after class) due to narrow walking and crowedness.



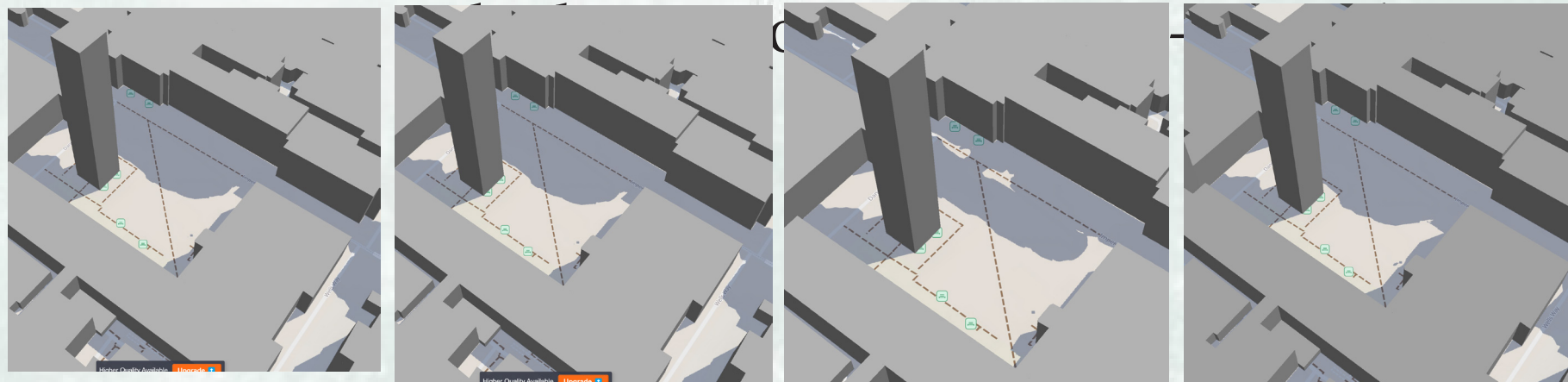
Good Weather

When weather is good, the grassland is **relatively crowded** with people sitting on the grassland enjoying the **sunshine**, chatting with friends under the sunshine. A student **dances with a ribbon** everytime at afternoon if the weather is good. Occasionally there will be a **parade or some visitors**. **Club fair** or activities are constantly held around the grassland.



Shadows Around the Year -- Simulation

Shadows create by sunshine in February, April, June and October at 14:00: People sit on grass when sunshine cover that area.



The **shaded area varies** throughout the year due to the **changing position of the sun**. In **February and October**, the shadows appear **longer**, **covering more** of the grassland. In contrast, during **April and June**, the shadows are **shorter**, **more sunlight** reaches the grassland

It is observed that areas that receive consistent sunlight will likely have **healthier and denser grass growth**. Conversely, areas that remain shaded for **long periods** may have **slower grass growth**.

Areas with **year-round shade** struggle with vegetation growth, while those with sun exposure support healthier grass.

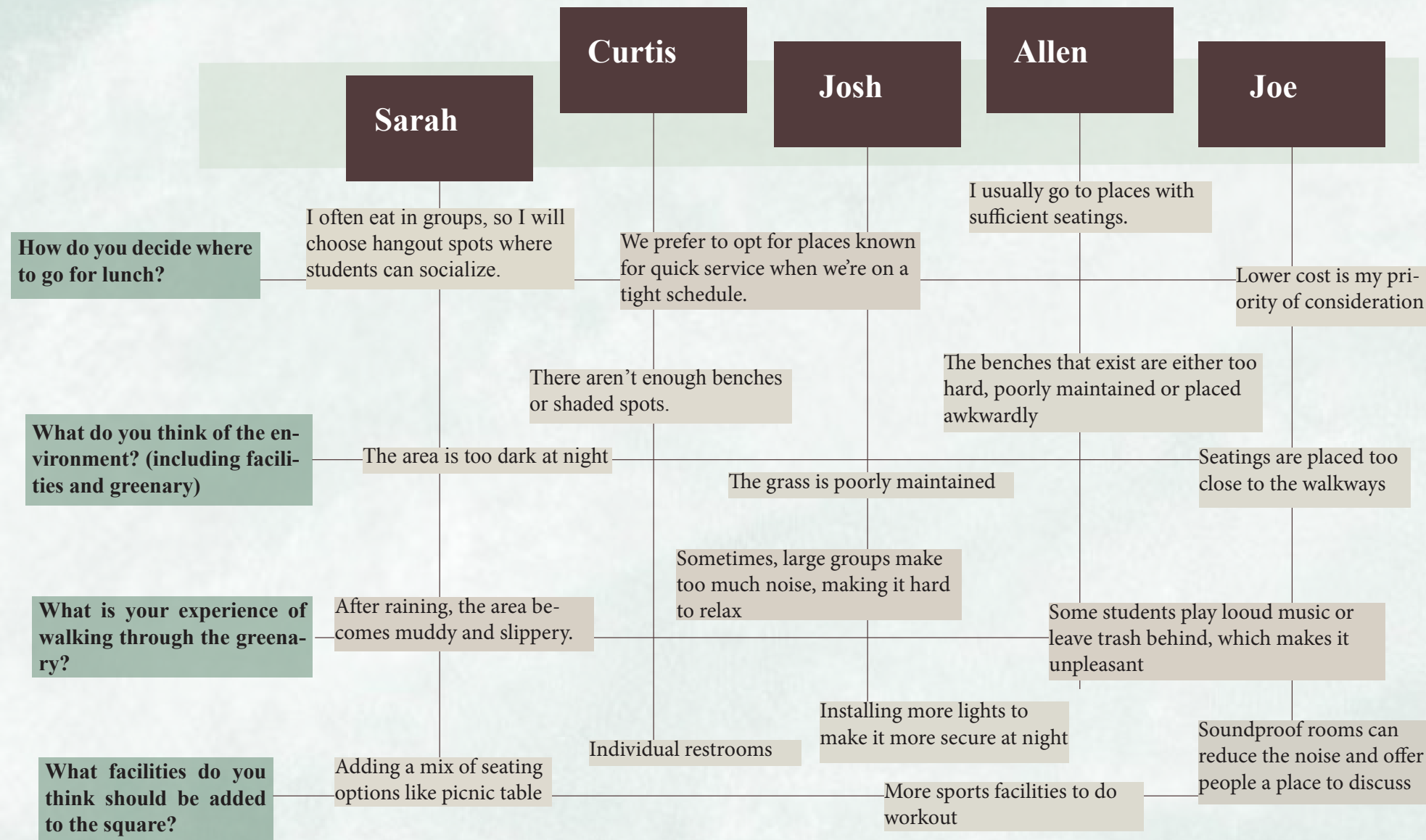
1st February

1st April

1st June

1st October

Interviews



Co-design Workshop



Opportunities Summary

Students

- Improved Aesthetic
- Sufficient Seating
- Food supply at night time
- More sport space

Staffs:

- More effectively used plaza
- School culture linked- promotion of school spirit

Maintenance Staff:

- Convenient and connected facilities
- Mental&Physical health can benefit from facilities



Professor Peter Childs

Professorial Lead in Design Engineering
Dyson School of Design Engineering - Faculty of Engineering

- The plaza was ineffectively used over the **past 17 years- few activities** were held there
- The plaza may have **potential constraints** which can affect the future re-design
- The plaza can be used to link the **school culture (hardworking)** as a showcase to the **visitors** for school promotion
- the current **showcase is a failure**, is not linked to hardworking.



Maintenance Staff

Maintenance staff of Imperial who is prioritising a new design for Dangoor Plaza and Queen's Lawn

- Current designs are **bad and inconvenient** for the students
- Facilities should be **connected** to improve **convenience**
- The current designs like the bench outside the library is **badly-designed** and **inconvenient- facilities** should save user time and improve user experience
- The plaza should be beneficial for both **mental and physical health** of the students and **keep them safe**

User Journey

User Mapping

An experience mapping was drawn based on a user’s emotion across the day. The curves one the right were drawn based on the current user experience (the light green) and the potential user experience with the improved facilities around Queen’s Lawn (darker green). Below are explanations of the different times of the day.



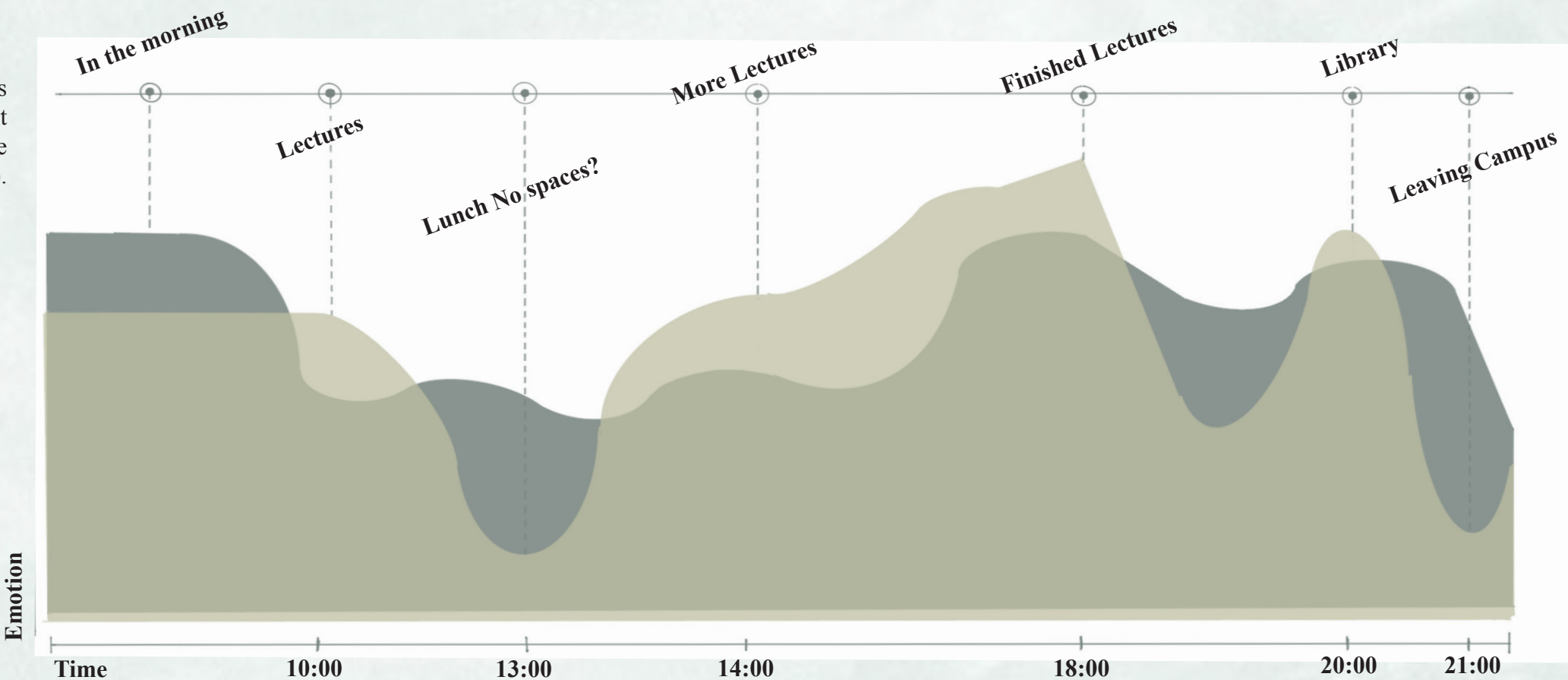
In the Morning : Some may want to go for a walk, but Queen’s Lawn is very empty, without any facilities.

Lectures: Normally not exciting and looking forward to go to.

Lunch: Trying to find a seat... but it is packed with people!

Library: Before finding a place, running around campus trying to find a place to work....

Leaving Campus: Leaving the library, no light, walking across the Queen’s Lawn lowers my mood.



Student Primary User

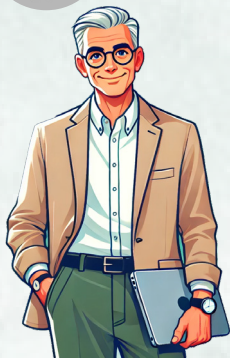
- 18 years old 1st year design engineering student
- a first-year design engineering student who spends long hours on campus, often study late to evening

- Desire a serene and visually appealing environment to relax between study session
- Seek for some sport facilities to refresh herself after long time of study.
- Struggle with insufficient seating
- Find it challenging to study after 7 PM because there are no available snacks or drinks provided to help her stay energetic.

Frequency



Satisfaction



Staff Secondary User

- a 50-year-old Environmental engineering professor who has been teaching in Imperial for over a decade
- Enjoy informal meetings with students in natural settings.
- Hold a Ph.D in environmental & sustainability

- Prefer to hold outdoor lectures or discussions but finds the campus grassland unsuitable due to lack of seatings and poor maintenance

- Finds it difficult to encourage students to appreciate the space when it is not well-maintained

Frequency



Satisfaction



Ribbon Player Extreme User

- a 2nd-year student who loves ribbon dancing.
- Practices two times a week, using the campus grassland as her primary space
- Prefer dancing on the grassland when the weather is good.

- require a open and appealing area for practice but find weather as a limitation
- Struggles with inconsistent ground surfaces that can make dancing difficult, especially after rain when the grass becomes muddy
- aims to collaborate with the university on campus events where dancers and other artists can showcase their talents in a structured manner

Frequency



Satisfaction



Visitors Extreme Non-user

- A 50-year-old retired accountant who lives adjacent to the university.
- He sometimes visits the campus to have lunch at the cafeteria, enjoying the social atmosphere and student energy, but he completely avoids the grassland

- Prefers indoor or structured seating areas for comfort.
- Find the grassland uncomfortable for sitting or walking, particularly after rain when it gets muddy
- wants more covered and well-maintained seating areas near the cafeteria

Frequency



Satisfaction



Insights

Users seek a more **aesthetically pleasing** environment for relaxation, but the **poorly maintained** grass and **lack of flowers** make it dull. Rain renders wooden benches unusable and the ground muddy, limiting use to sunny days.

Insufficient seating forces students to stand, sit on the floor, or seek alternative spaces, **discouraging** them from staying on campus and limiting their learning and engagement opportunities. When students do find a bench, the **narrow placement** and **uncomfortable orientation** make resting unpleasant. As a result, benches are rarely used, further reducing the functionality of public seating areas. Moreover, **shading of sunshine** should be considered for **arrangement** of the facilities for different activities.

Students need more **on-campus services**, especially for those working late in the library, as there are **no places to get snacks or drinks after 19:00** to replenish energy. Additionally, they seek more **sports spaces** since Ethos Gym is often **crowded**, and good weather provides opportunities for outdoor activities. However, they are concerned about **potential damage** to the grass and environment caused by sports activities, which affects their willingness to use open spaces for exercise.

Users want better **connectivity** between facilities to enhance **convenience**, allowing students to access them more easily. However, they are concerned about the current **poor** and **impractical** designs. They also seek a space that promotes both **mental and physical well-being** while ensuring **safety**, as these are crucial for students. Additionally, they want the area to reflect the school’s **hardworking culture** as a way to showcase it to visitors, but they feel the current design fails to achieve this.

The maintenance staff wants a **sustainable** area as the area contains a high amount of greenery which is **eco-friendly** and allows students to relax and release their stresses. However, the staff concerns about the **potential damage to the greenery** due to future activities.

HMW Questions



Aesthetic & Weather free & Interaction: HMW create more **aesthetically pleasing** and **functional spaces** on campus, where students can relax, study, and engage with the environment, **interacting** with others, ensuring comfort and convenience **in all weather conditions**?



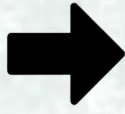
Capacity & Arrangement: HMW redesign seating areas and public spaces on campus to increase **capacity**, improve **comfort**, and improve **arrangement** of facilities to encourage students to spend more time engaging in learning and activities?



Food & Comfort & Sports: HMW provide more accessible **food options** and **sports spaces** on campus, especially **after 7pm**, while balancing the preservation of the environment and promoting student well-being?



Convenience & Safety & Culture: HMW enhance **connectivity**, **convenience**, and ensure **safety** across the grassland, keeping seamless movement between facilities and creating areas that reflect the **school’s culture** and support both physical and mental health?



Sustainability: HMW create a **sustainable green space** that allows students to relax and enjoy nature while ensuring **minimal damage** to the **greenery** and reducing maintenance challenges for staff?

Ideation Overview

Initial Ideation from research

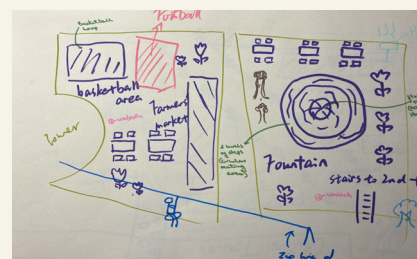
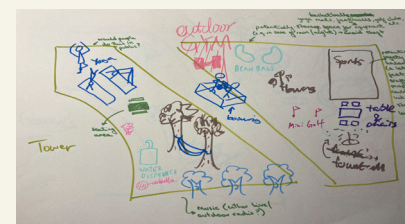
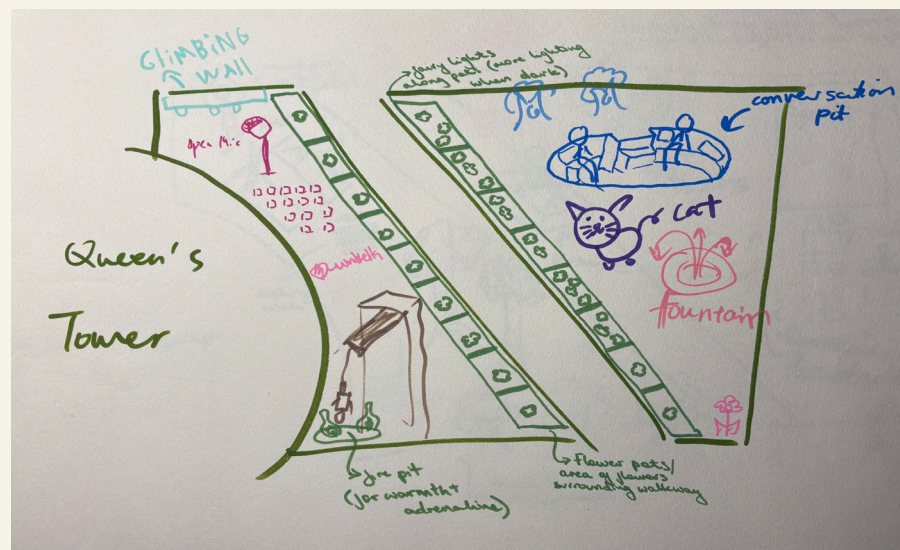
- Based on **everyday observation** and the needs as a student
- Create both **physical and digital** ideas
- Focused on **user experience** and needs to solve the problem

Co-design activity

Gathered students from **different courses** at Imperial (age group of 19-20) to work with 5 group members, **10 participants** in total.

The activity contains of 3 stages:

1. **Users experience and survey responses** and their current concerns are collected and analysed.
2. Ask users about their **desired design** of the Dangoor Plaza and conduct a discussion.
3. Deliver out **papers** to the participants and ask them to **draw out** their dream design of the plaza



Idea evaluation and selection

Table for marking and selection

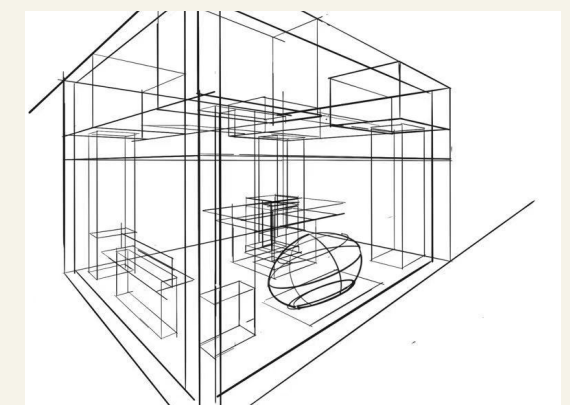
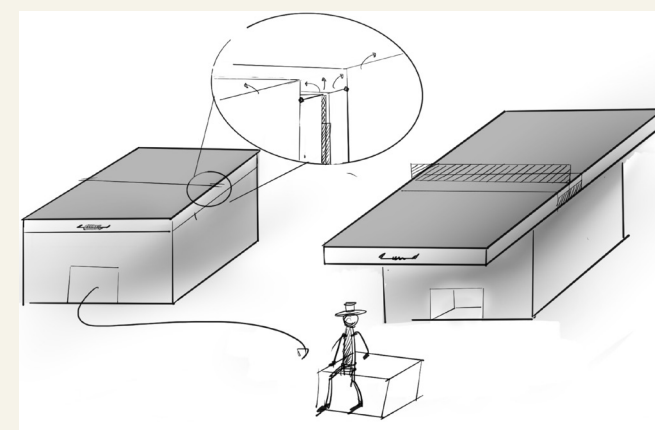
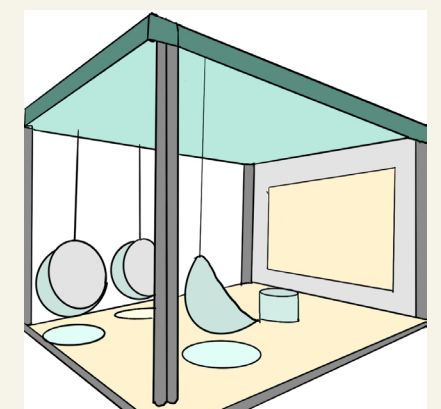
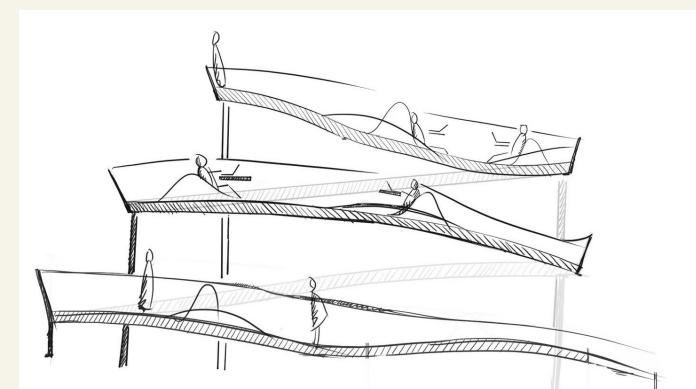
- marking **15 ideas** from different aspects including **desirability, feasibility, viability, aesthetic, and connectivity & convenience**

Venn Diagrams for important criterias: desirability, feasibility, viability.

- Desirability, feasibility and viability of all ideas is compared through Venn Diagrams
- Ideas nearest to the centre of the Venn Diagram **performs the best** for the feature

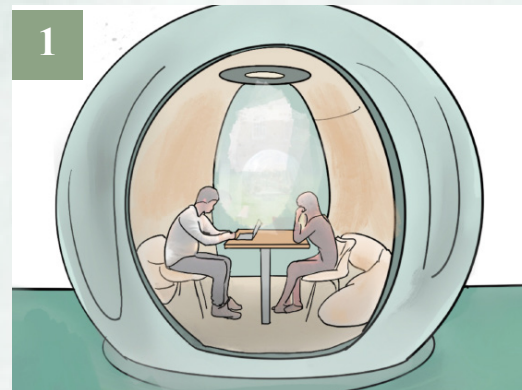
Concept Development

- Based on concepts of **ideas prioritised**, other ideas and details are added to form **3 complete models** for urban design in Queen's Lawn
- **Detailed sketches** of each component to analyse each complete idea
- Complete user testing for **feedback**
- **Improvements** made according to users' feedback
- Develop the concepts to produce **final designs**
- Final three ideas are modelled by **CAD & Minecraft**



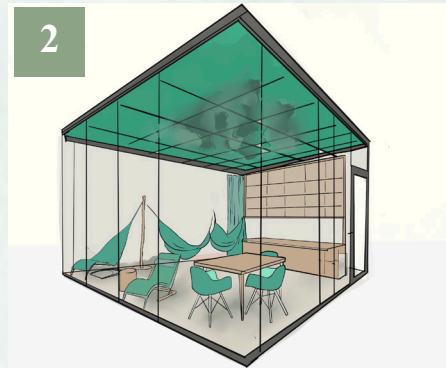
15 Ideas

15 ideas are generated according to 5 HMW questions each corresponding to several crucial aspects.



Bubble Study Pod (Aesthetic & Capacity)

An **isolated transparent** room designed for **group-study**, while keeping the space isolated.



Glass House (Seating & Weather Independent & Comfort)

A house made out of **glasses**, allowing students to study and resting while enjoying the **sunshine**.



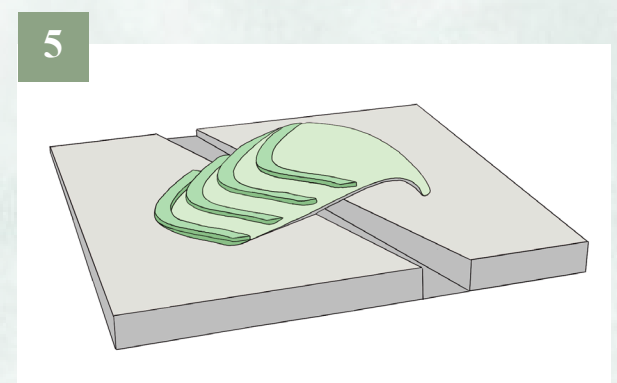
Outdoor lecture theatre (Seating & Culture)

An **outdoor theatre**, allowing lectures to be held outdoor and promoting the **hardworking** school culture.



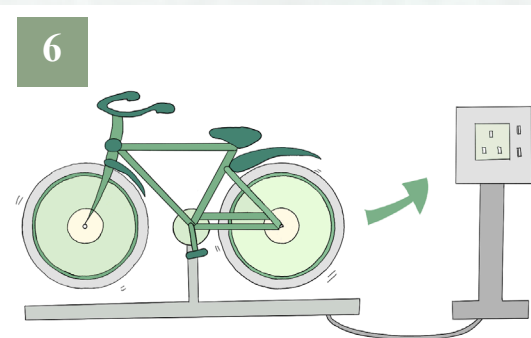
Tree Spiral Staircase (Aesthetic & Capacity)

A **spiral staircase** around a tree, designed to increase the **seating** and allow students to enjoy the **view**.



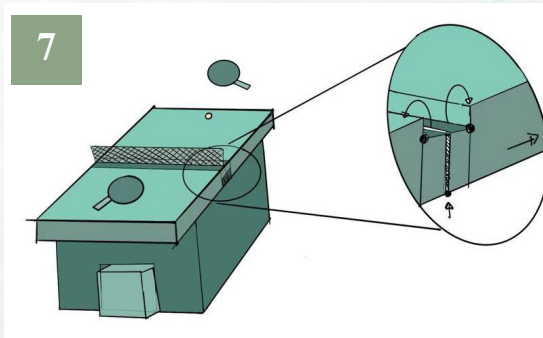
Lawn Connection Bridge (Aesthetic & Capacity)

A bridge **across** the two lawns, providing **convenience** and more **seatings**.



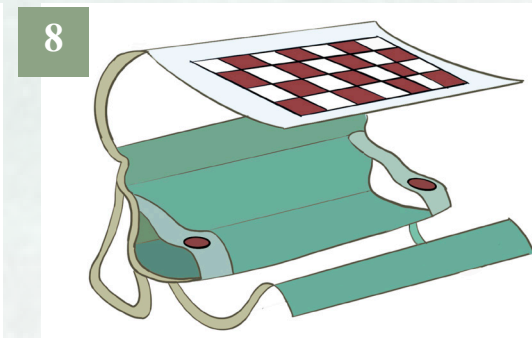
Bike Charger (Sports & Sustainability)

A bike that both satisfies the needs of **sports** and is able to provide **energy for charging** sockets.



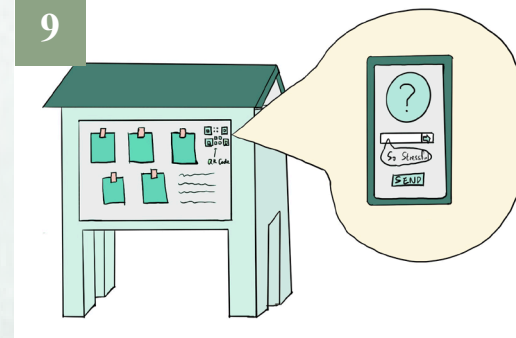
Transformable Table for table tennis (Sports & Interaction)

Can be used as a table for **self-study or eating**, and can be transformed for playing table tennis.



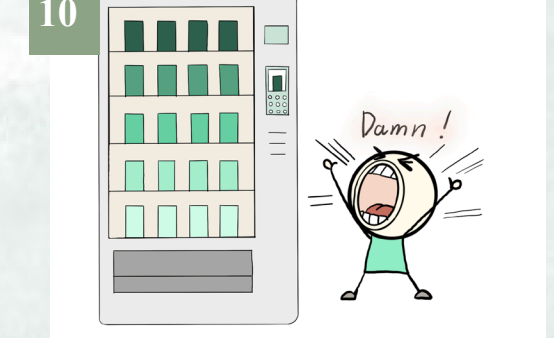
Multi-functional bench (Comfort & Convenience & Sustainability)

A multi-functional bench that combines functions including **foot-resting**, **wireless charging panel** and etc, providing convenience for student



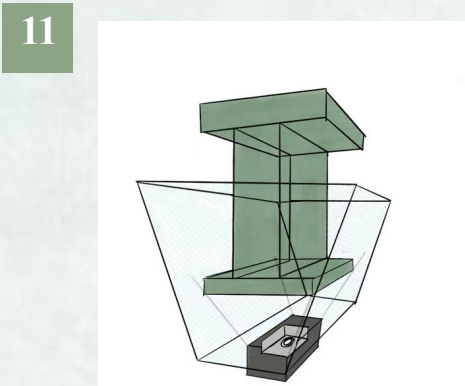
Communication Board (Interaction)

A board designed to improve **interaction**, allowing students to **leave/respond** any questions (communicating through the board).



Stress-releasing Vending Machine (Food & Stress)

A vending machine designed to help students release their stress- the louder they scream, the more **energetic the drink** provided.



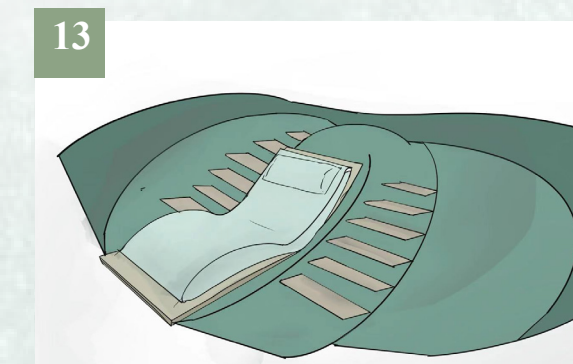
Hologram Project (Aesthetic & Culture)

A **hologram projection** that is designed to display things which improve the appearance and promote **school culture**.



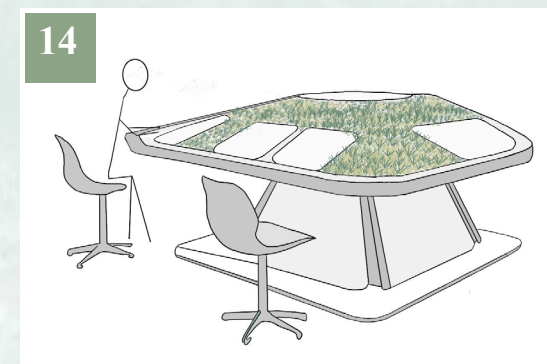
Touchable Screen Walkway (Aesthetic & Culture)

A walkway with a screen under to **display images** and provide **lighting** for the students. It will be lightened when someone steps on it which provide lighting in the evening.



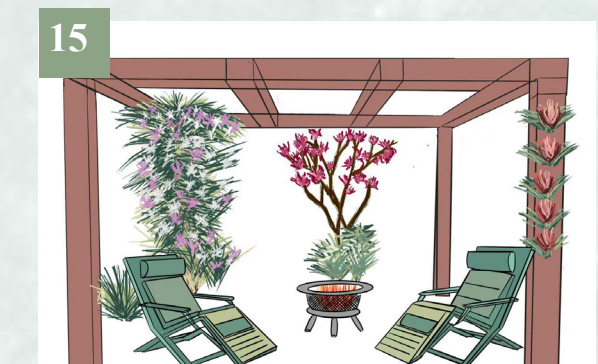
Grassland Embedded Lounge (Capacity & Comfort)

Ergonomic Lounge embedded to the slope of the grassland which should be further built providing **comfort** and **seats** for students.



Bioenergy Study Tables (Capacity & Sustainability)

Desktop implanted with a **special plant** that converts the **energy** generated by **photosynthesis** into USB **charging**, allowing students to use electronic devices in a green environment.



Blossom Heaven (Capacity & Comfort & Aesthetic)

Provide a peaceful spot where students can **relax, read, or rest** under a wooden **shelter** with **flowers** and comfy foldable lounge chairs.

Ideation & Selection

Ideation Techniques

Sport

- **Fact:** Students complaining about **insufficient spaces** of workout in campus
- **Opportunity:** To **combine sport** space with study **space-transformable**

Study & Capacity

- **Fact:** Students cannot find seats in **library**, especially between **1PM to 4PM**
- **Opportunity:** Add more study space on the grassland which is **weather-independent**

Other

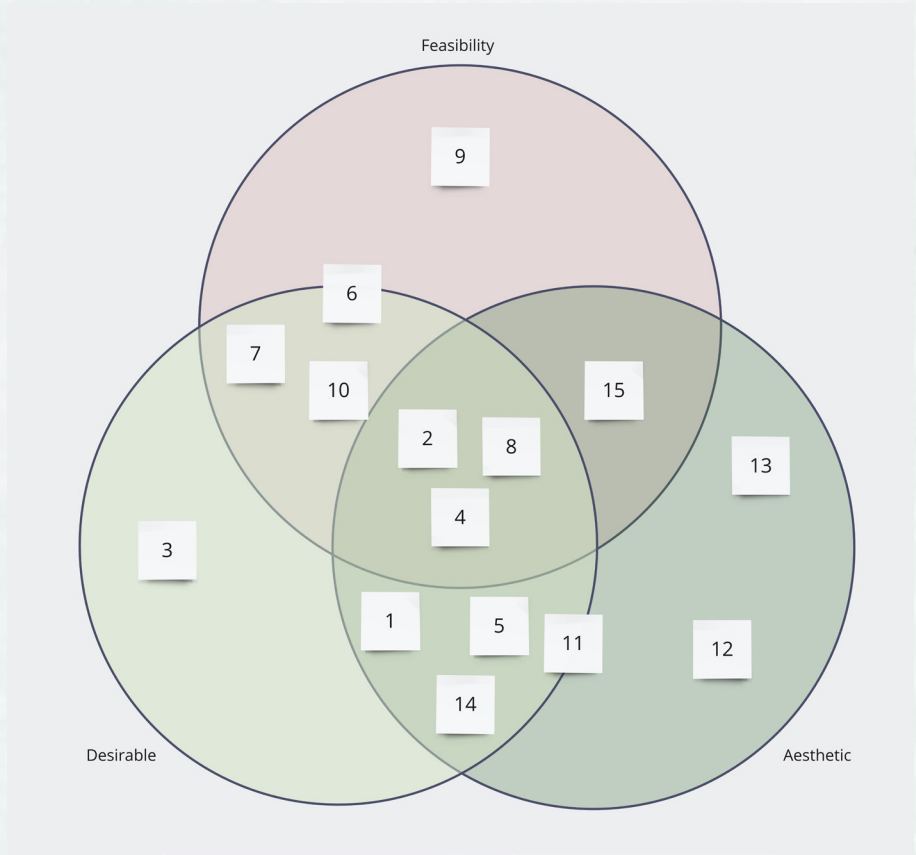
- **Fact:** There is **lack of food** supply at night, most facilities are **inconvenient** and cannot promote **school culture**.
- **Opportunity:** Combine **sustainable technology** with the facilities to achieve the purposes.

Evaluation of 15 ideas -- User Rating

The targeted **primary users**, which are the students, rated the 15 ideas from 1 to 5 taking into account 5 factors: **desirability, feasibility, aesthetic, capacity, and connectivity & convenience**. The surveyed group contained 10 students aging from 18-20, from courses including Chemistry, Design Engineering and Mathematics.

Criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Feasibility	4	5	2	3	3	4	4	5	5	4	3	2	5	2	5
Aesthetic	5	5	4	5	5	3	3	3	3	2	5	5	5	5	4
Capacity	3	5	5	5	5	1	2	1	1	2	2	4	5	3	3
Connectivity & Convenience	2	5	5	4	4	3	5	3	3	5	3	3	2	4	4
Desirability	3	5	3	3	3	3	5	3	3	5	3	5	3	3	4
Total	17	25	17	20	20	14	19	22	15	18	16	19	20	17	20

Venn Diagrams



A **Venn diagram** is generated to compare the ideas in three aspects:

1. **Feasibility**- How **practical** is the design
2. **Desirability**- Do users **want** the product
3. **Aesthetic**- How good the product **looks**

By analysing and selecting ideas, the three ideas, **idea 2, idea 4 and idea 8** are selected for further analysis.

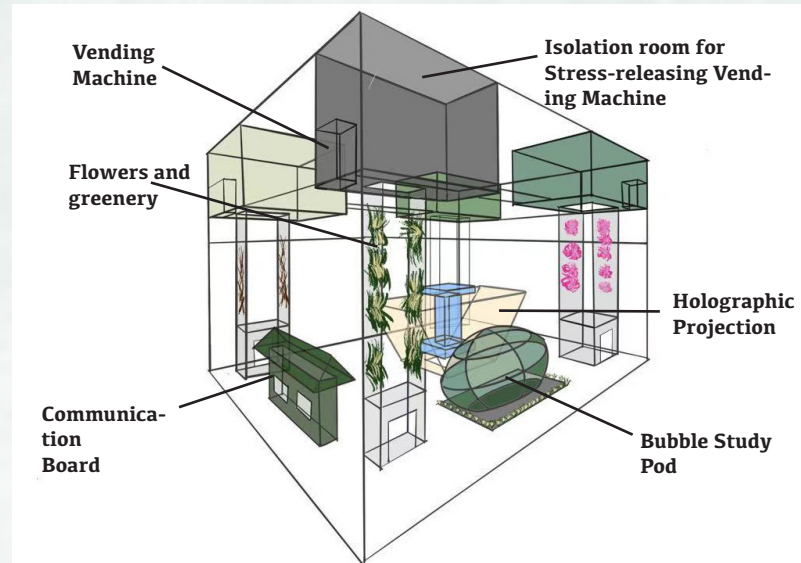
Idea 2 stands for the **glass house**, which is the best idea with highest score, followed by **idea 8, the multi-functional bench**. The third place is **idea 4, 5, 13, and 15 (Tree spiral staircase, Lawn Bridge, Grassland Lounge, and Blossom Heaven)**.

The selected ideas will be later analysed by **3D modelling** and **sketch drawing**. Continuous changes will be considered due to user feedback.

Concepts Development

Three Ideas of *BEST* Feasibility, Desirability, and Aesthetics are Selected for Further Developments and Improvements.

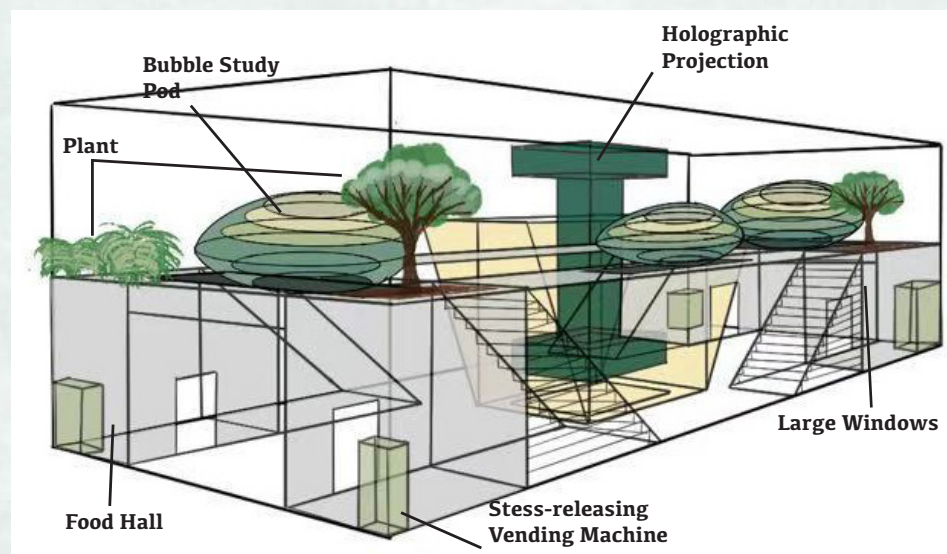
IDEA 2 Glass House



User Feedback

- Might be **too large** for the Queen's lawn, one floor should be enough
- **Lift** might be difficult to realise- digging
- Study pods **not large** enough
- Lack of **ventilation** and plants
- Lack of **charging** socket

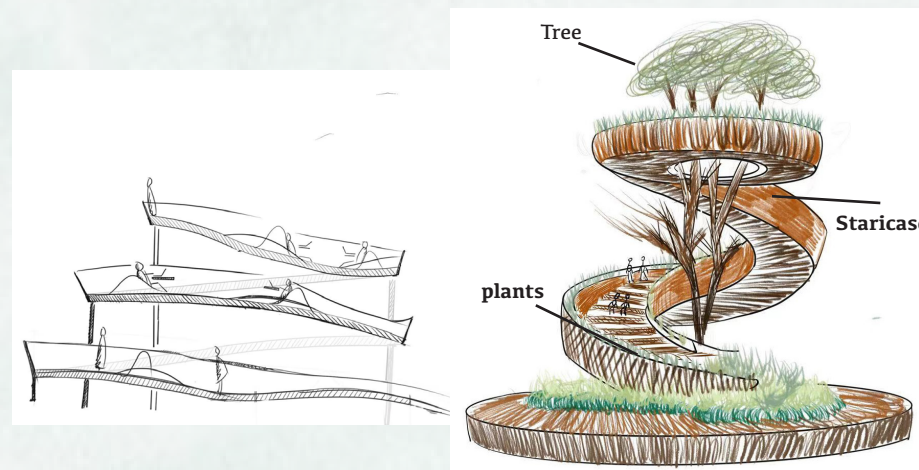
Sketch after Improvements



All lifts are changed to **stairs** and the **holographic projection** of sign of "I" is in the middle. Trees and **greenery** are planted on the ground. Moreover, shouting rooms are with **vending machines** and **sound-absorbing** sponge to release stress. Ventilation improves by **windows**.

IDEA 4 Tree Spiral Staircase

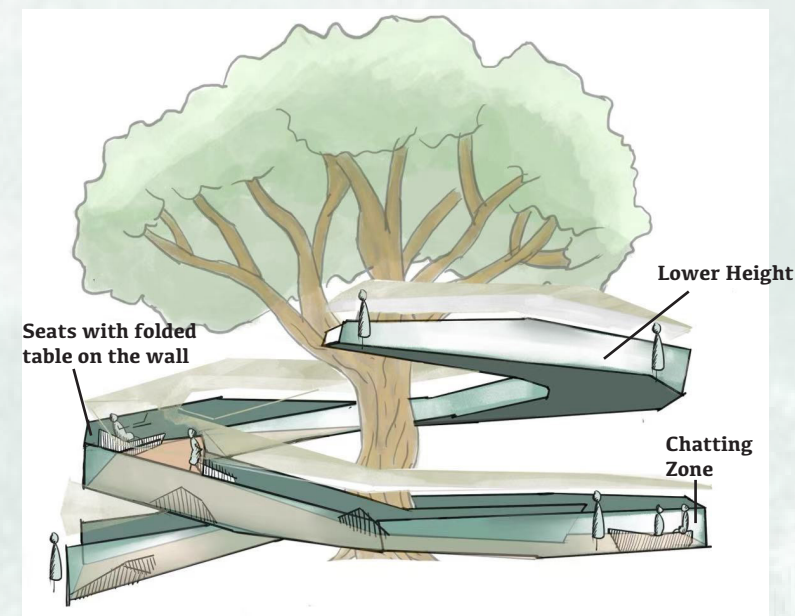
Contain Staircase for seatings and provide a good view of the landscape



User Feedback

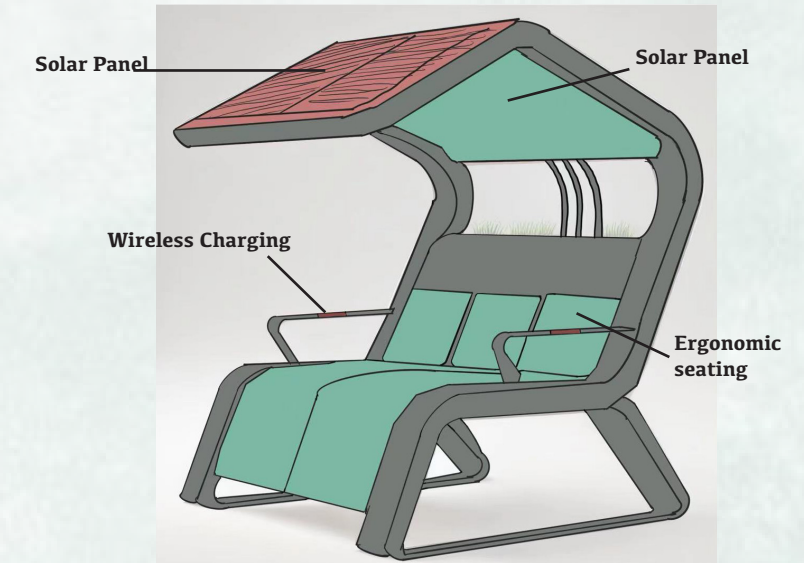
- Structure might be **too high** which might be scary and there will be some **safety issues**.
- **Actual seats** might be better than using natural staircase
- **Non-waterproof** staircase might cause some problems when it is raining

Sketch after Improvements



The whole spiral staircase is **lowered** due to safety issues. Additionally, cover of the staircase enables versatile utility for different **weather condition**. Moreover, **seats** are designed with favor of the height difference along the stairs.

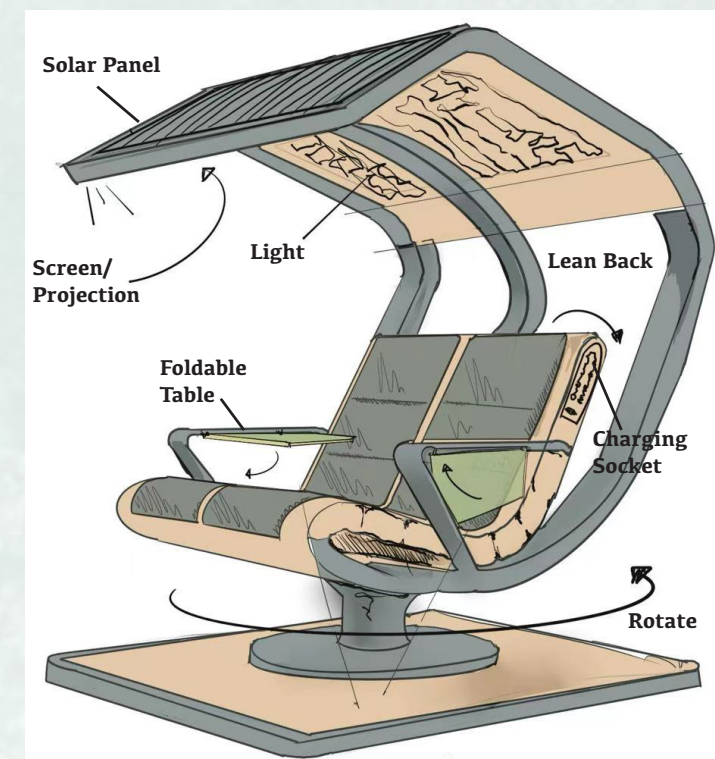
IDEA 8 Multi-functional Bench



User Feedback

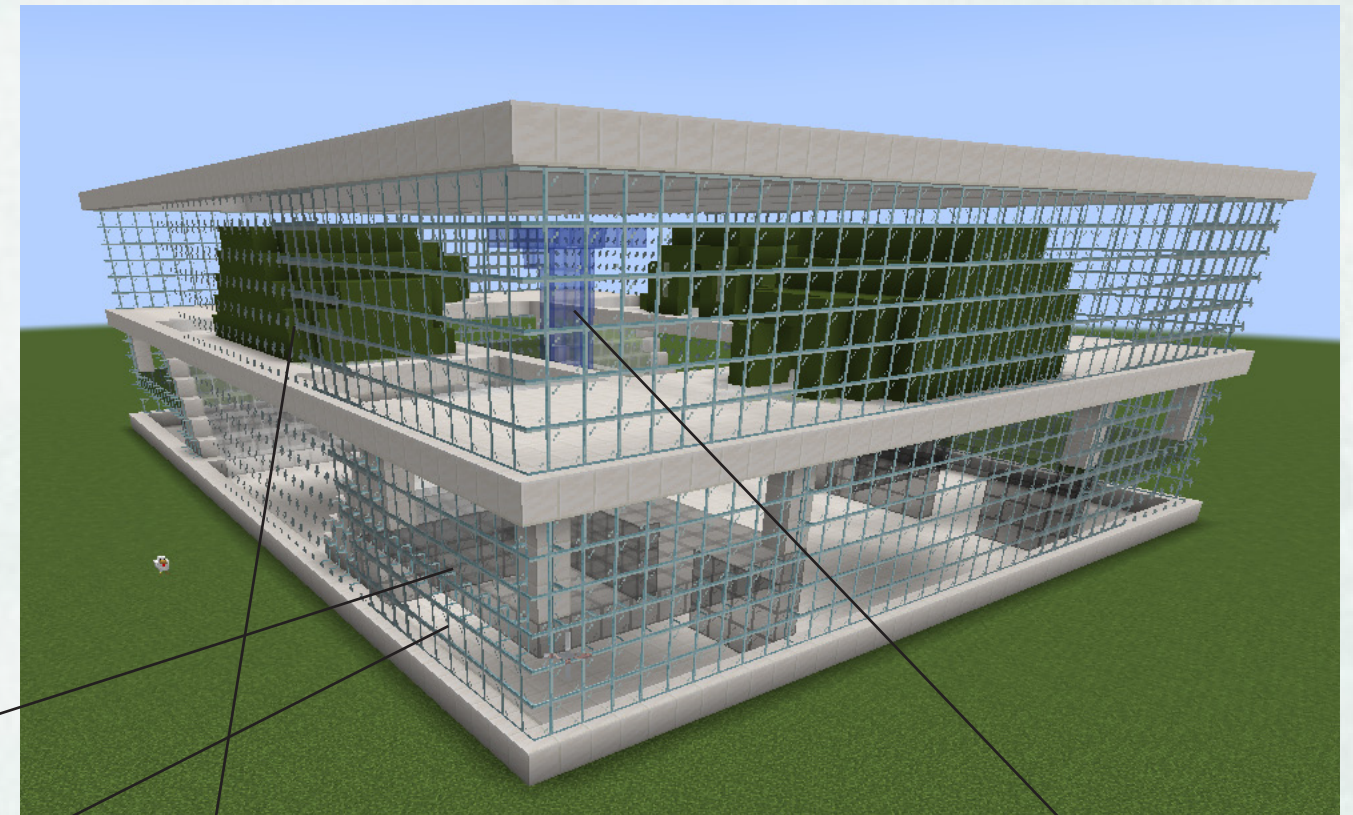
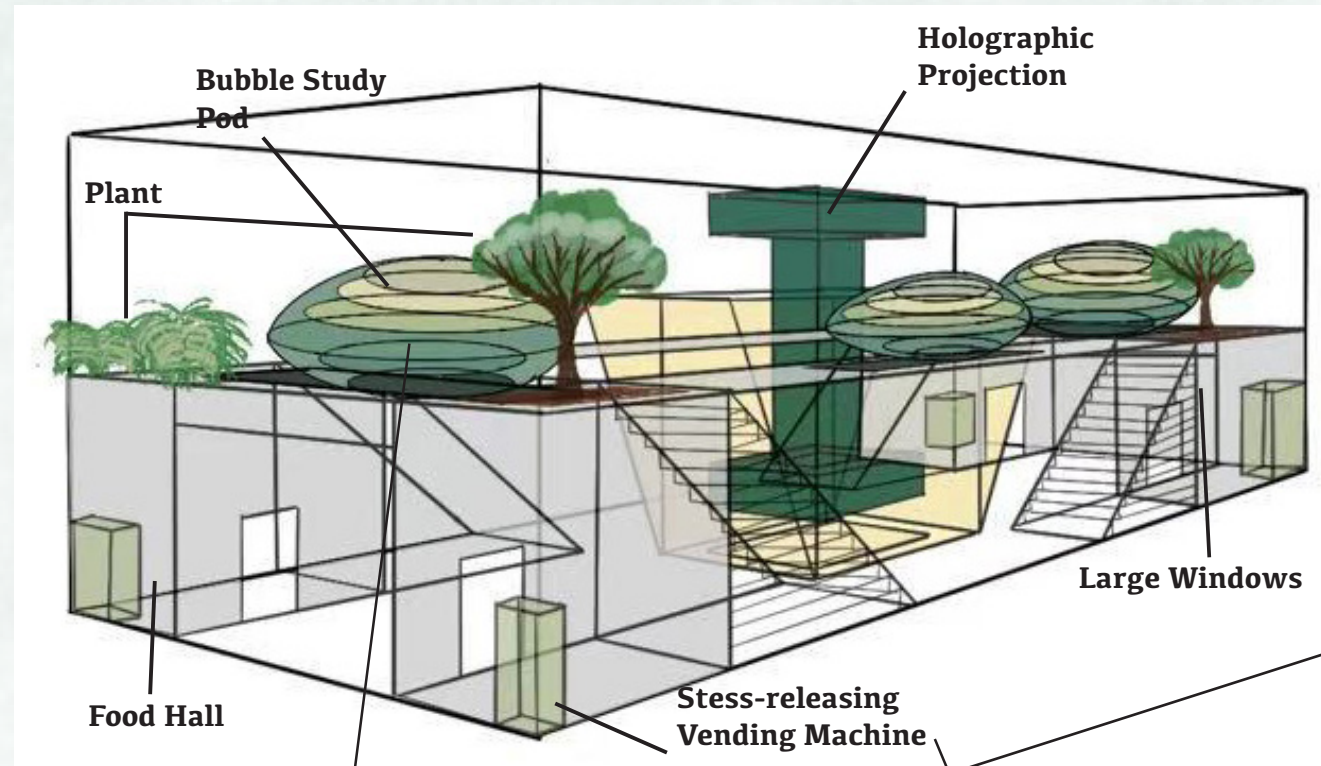
- Change of **direction** is limited
- **Appearance** can be improved
- Lack of **decoration**
- **Foldable** tables could provide more flexible functionalities for rest and study.

Sketch after Improvements

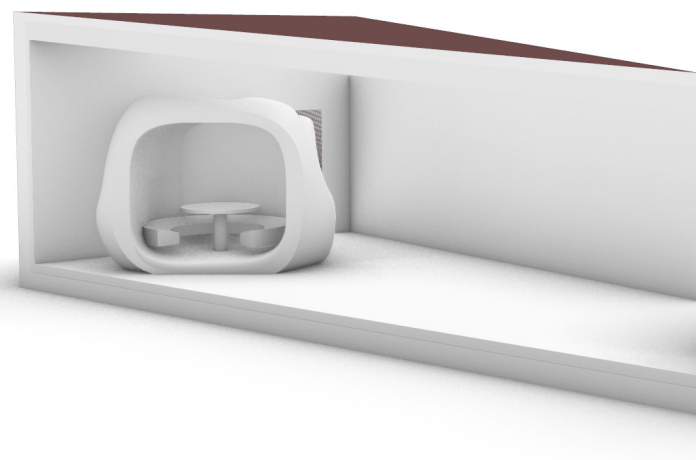


The bench could **rotate 360** for any direction (eg. always face the sun). The back of the bench could **lean back** for comfort. A **foldable table** is added to the bench for study. For sustainability, power collected by **solar panel** is used for **wireless charging** for phones and charging **socket** for other devices. Furthermore, a **screen** is added below the solar panel.

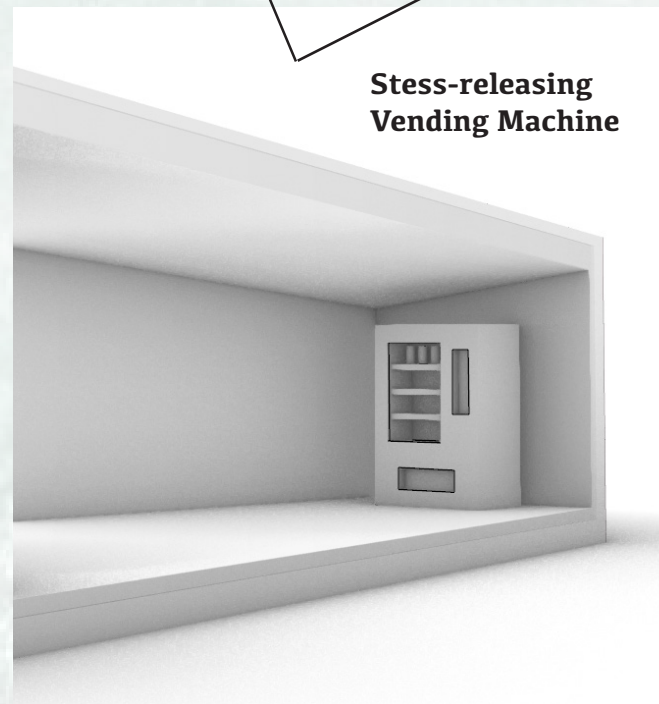
Architecture that provides great aesthetic with a holographic projection, self and group space for both eat, study and chatting with interesting devices and good view of greenery. It is available in all weather condition.



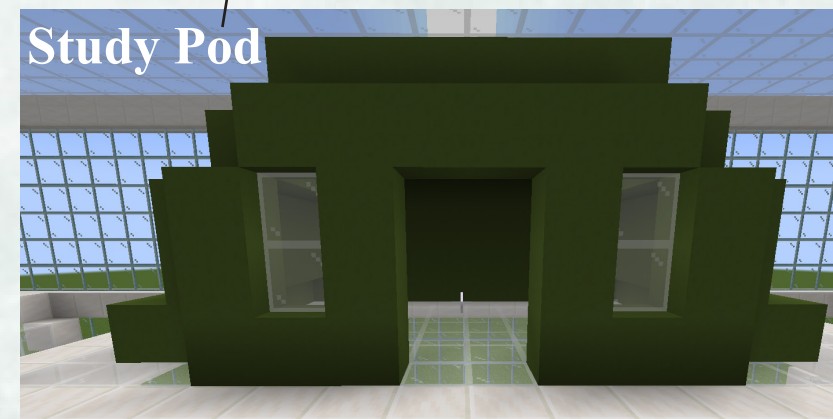
Bubble Study Pod



Stress-releasing Vending Machine



Study Pod



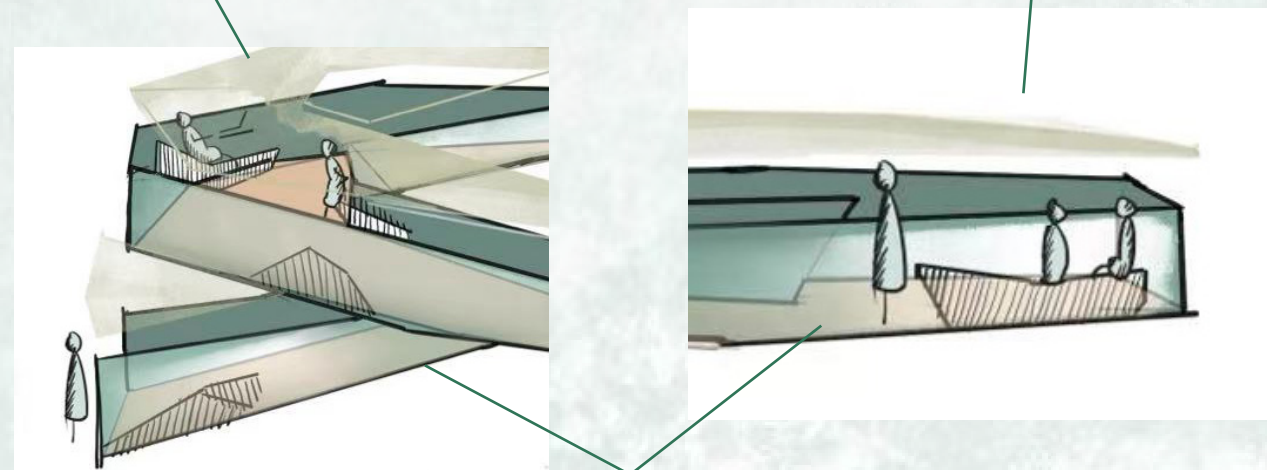
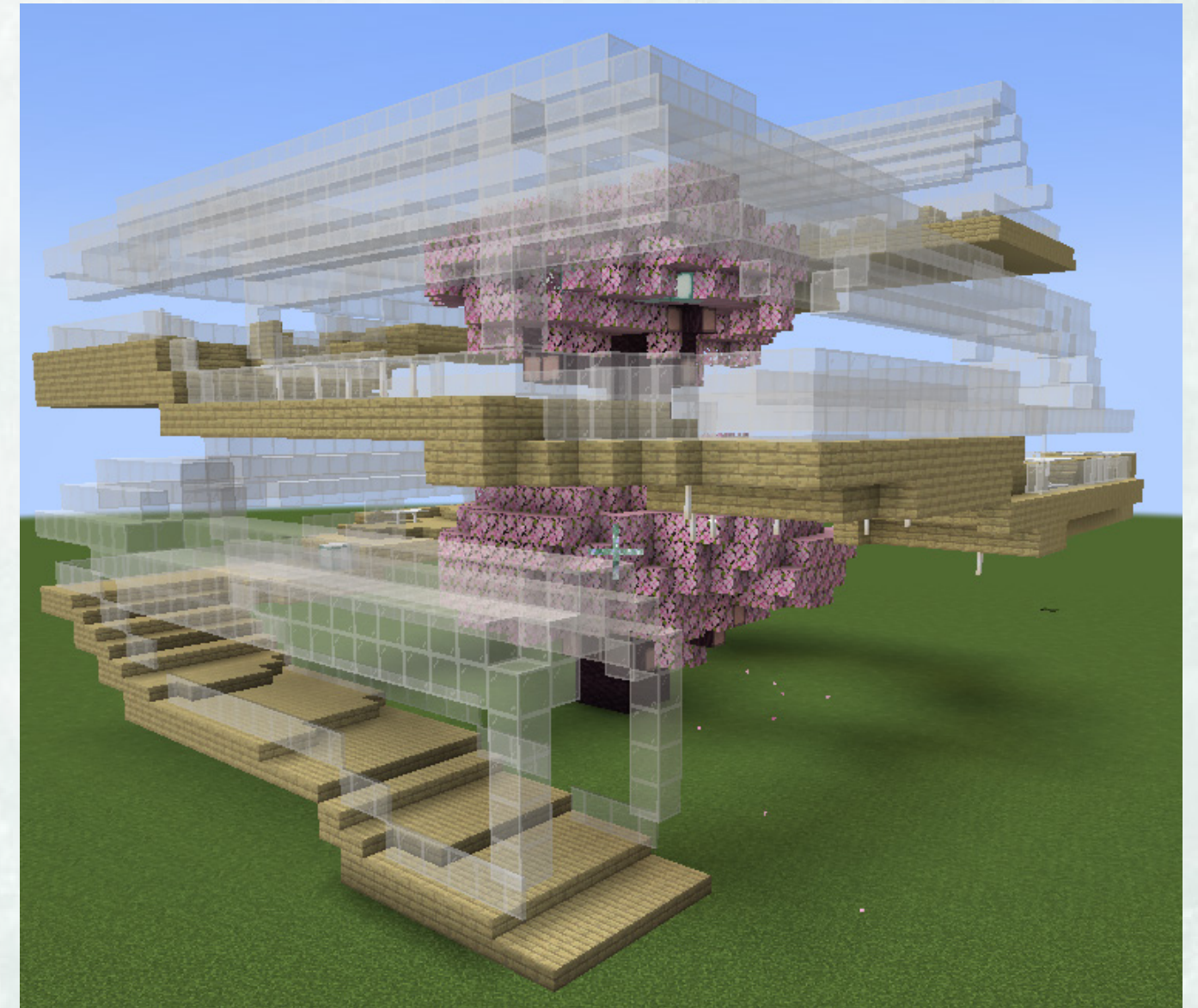
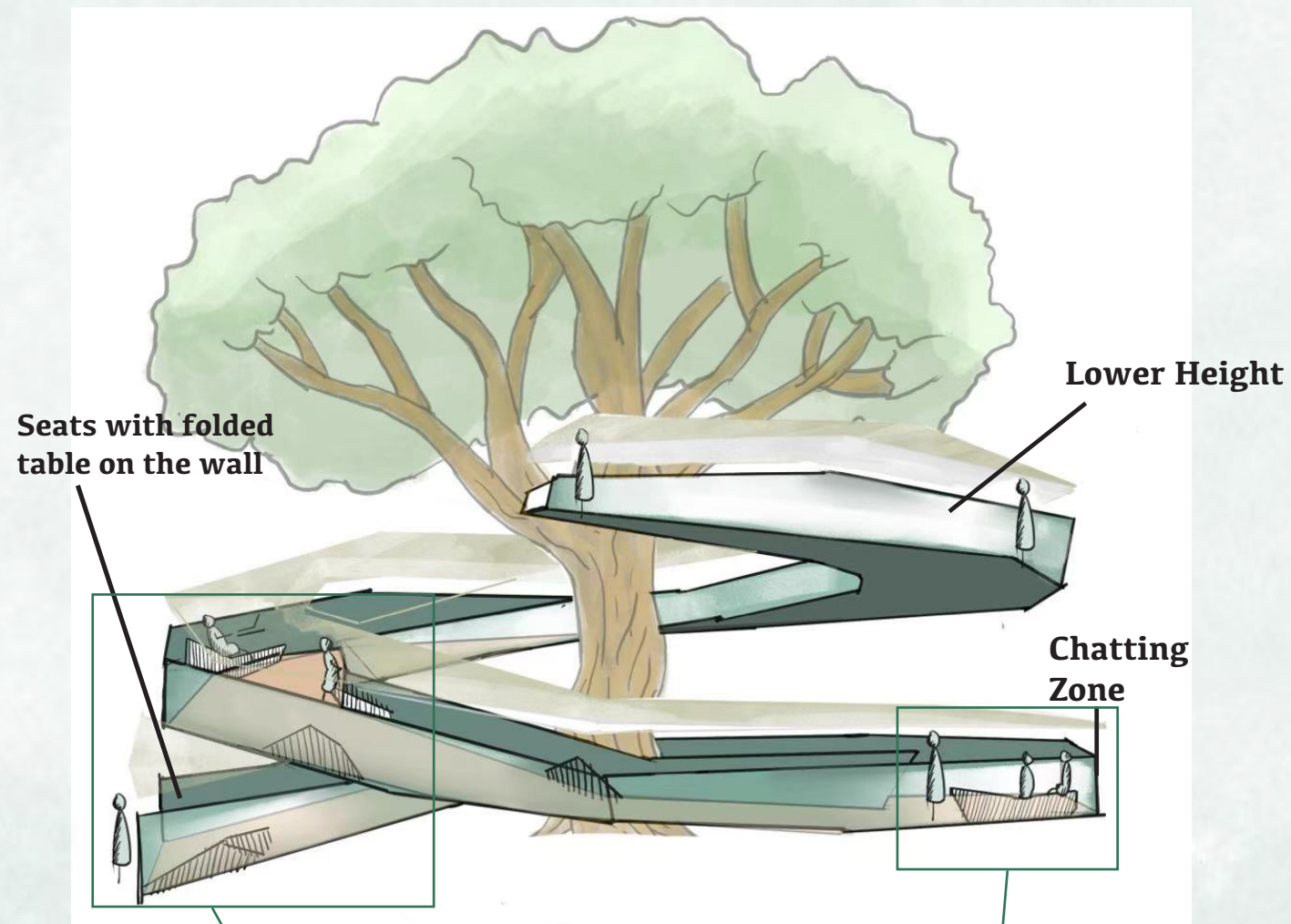
Holographic Projection



Inner View



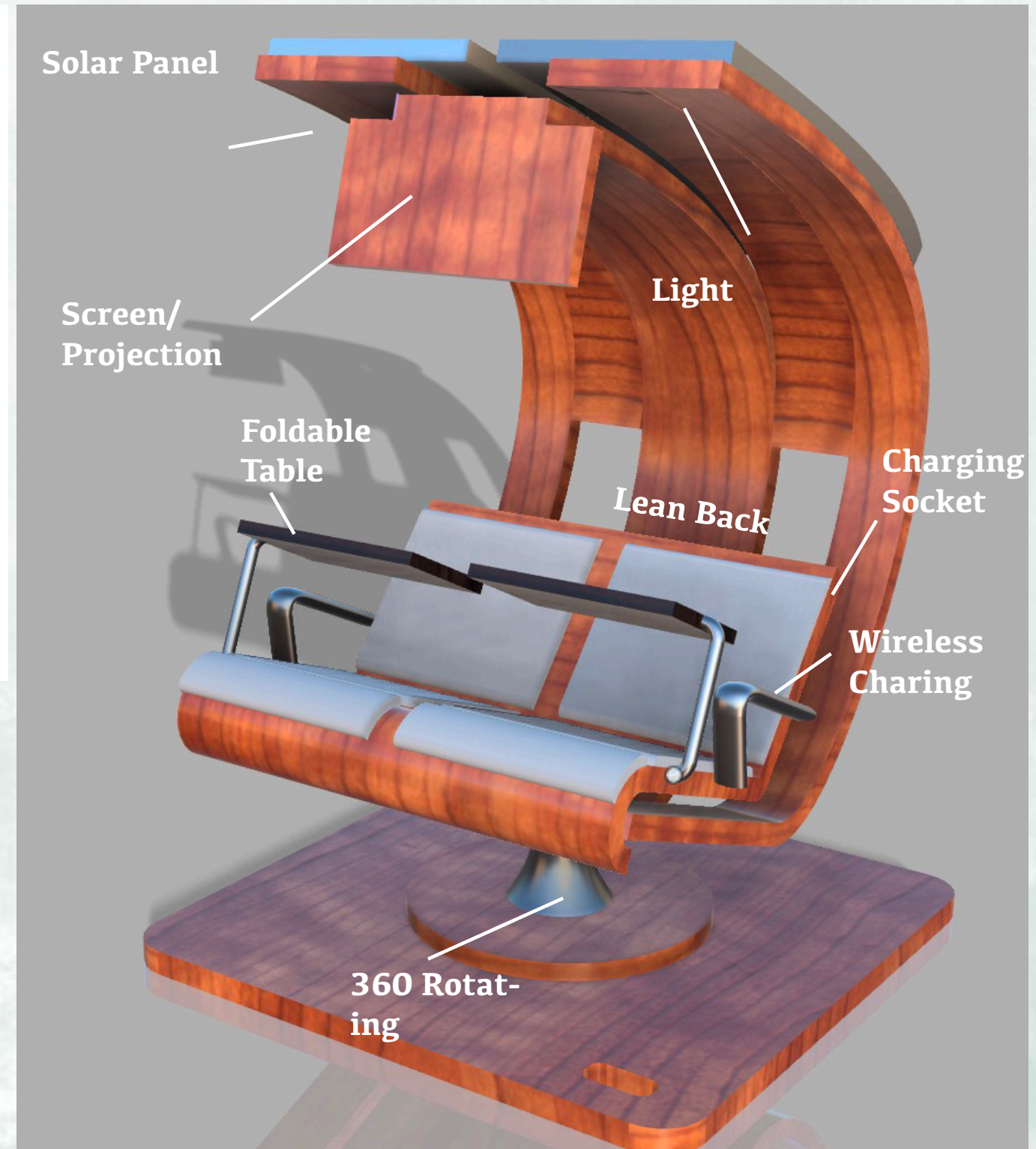
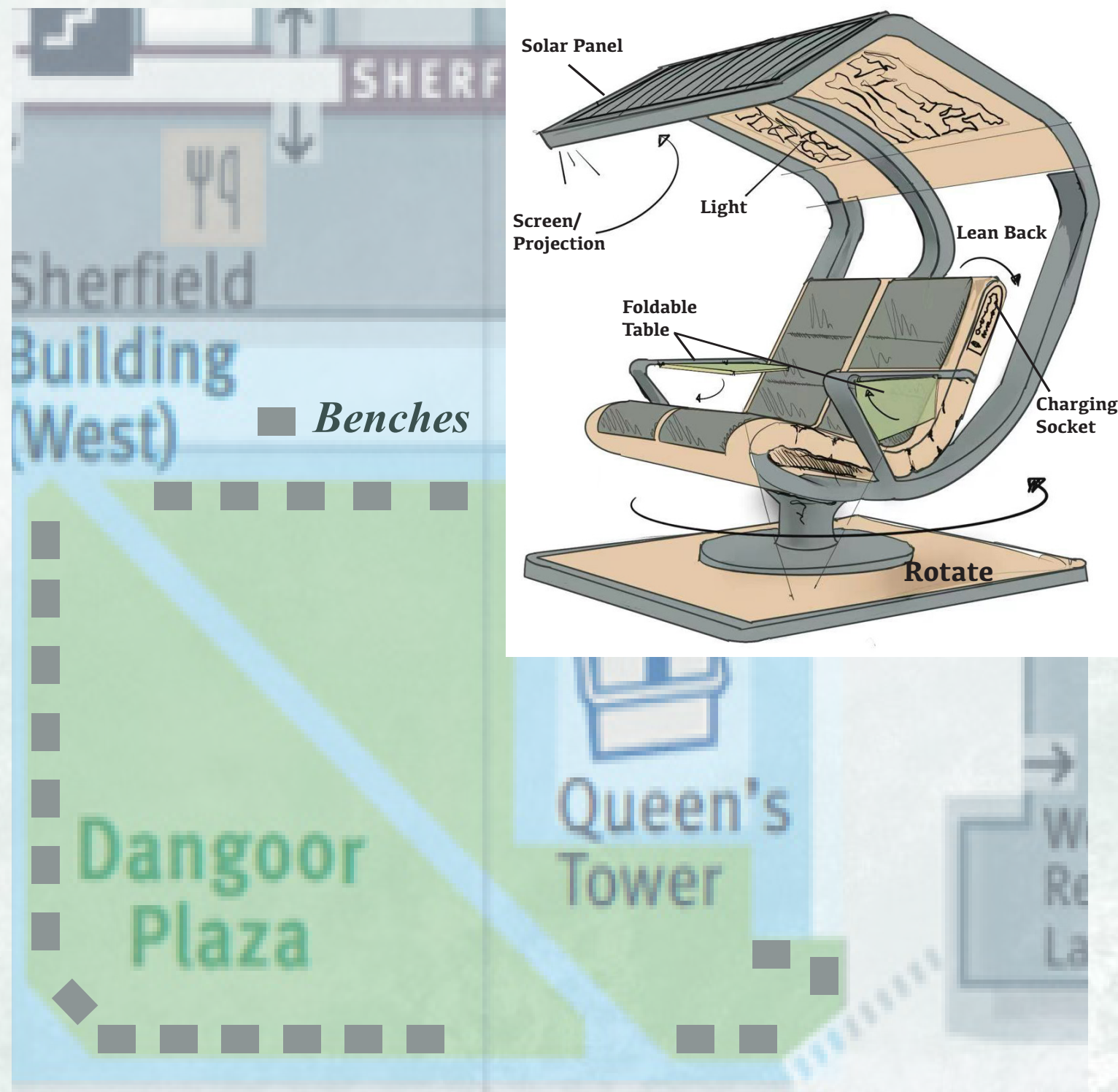
Staircase around the tree providing good view for refreshment after work. Seats along the hole staircase with tables offer a great atmosphere for study and help with the problem of limited seats. Waterproof architecture service in all weather conditions.



Seatings & Tables



Bench situated along the edge of the lawn with 20 of them. Solar panel is used for charging to ensure its sustainability and sheltering. It is design to be ergonomic for comfort purpose with foldable tables and screen for study. 360 rotation of the bench offers available view in all direction.

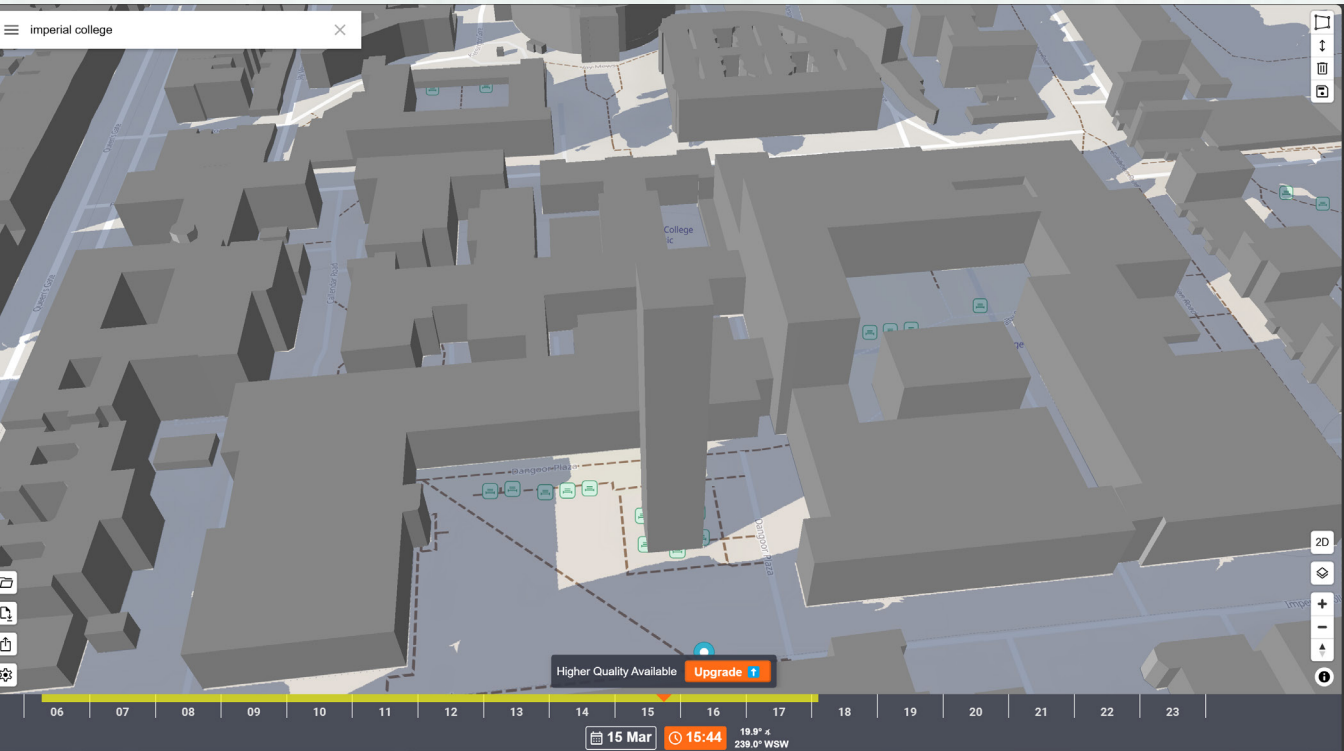


Appendix

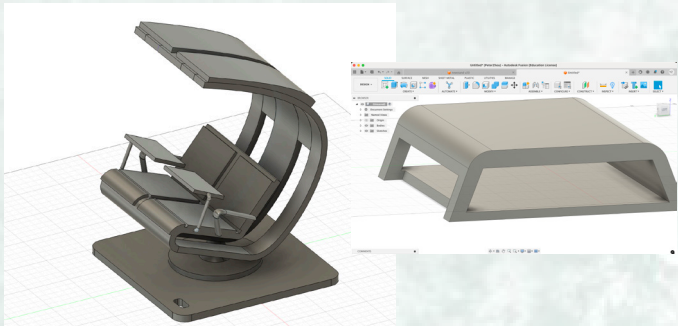
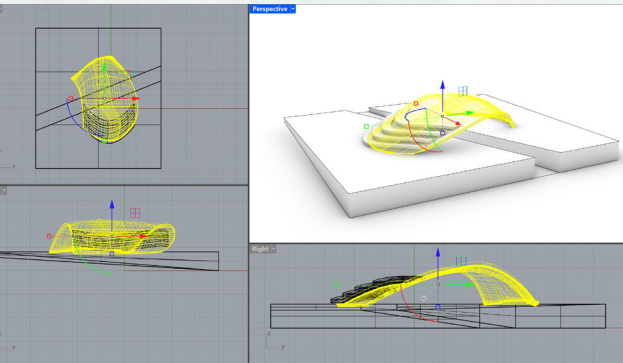
Sunshine and shadow simulation

Diagrams of simulation are created to simulate the sunshine and shadow across the year, helping to analyse the best position for the facilities.

<https://shademap.app/@51.5005,-0.1905,15z,1742071435079t,0b,0p,0m>



Modelling for ideation



Diagrams source

Tables and Venn diagrams: miro



Pie chart and bar chart: Python Coding

```
import matplotlib.pyplot as plt

# Data
labels = ["Have lunch", "Relax", "Socialising", "Passing by"]
responses = [2, 6, 0, 36] # Number of people
colors = ["#464F46", "#8B8E8B", "#5D645D", "#B8B8B8"]

# Create the pie chart
plt.figure(figsize=(6, 6))
plt.pie(responses, labels=[f'{label} ({count})' for label, count in zip(labels, responses)],
        colors=colors, startangle=140, autopct=lambda p: f'{p:.1f}%' if p > 0 else '')

# Title
plt.title("What is your primary reason for visiting this grassland? (44 responses)")
centre_circle = plt.Circle((0, 0), 0.65, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

# Show the chart
plt.show()

# Data
labels = ["SAF Building", "Classroom/ Department", "Dangoor Plaza", "Library Cafe", 'SCR', 'JCR']
responses = [11, 3, 4, 3, 14, 10] # Number of people
colors = ["#2F3A2F", "#8B8E8B", "#5D645D", "#464F46", '#B8B8B8', '#747974']

# Create the pie chart
plt.figure(figsize=(6, 6))
plt.pie(responses, labels=[f'{label} ({count})' for label, count in zip(labels, responses)],
        colors=colors, startangle=140, autopct=lambda p: f'{p:.1f}%' if p > 0 else '')

# Title
plt.title("Where do you usually have lunch? (44 responses)")
centre_circle = plt.Circle((0, 0), 0.65, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

# Show the chart
plt.show()
```

References

1. Map of Imperial College London showing the location of Queen's Lawn and Dangoor Plaza: <https://www.imperial.ac.uk/>
2. Pictures for personas: Generated using AI
3. Examples of sencondary research:
 - <https://www.whitecityinnovationdistrict.org.uk/new-public-garden-opens-in-white-city-innovation-district/>
 - <https://www.imperial.ac.uk/white-city-campus/whats-here/green-spaces/>
 - <https://www.imperial.ac.uk/white-city-campus/whats-here/green-spaces/>
 - <https://www.pinterest.com/pin/110127153358879321/>
4. Softwares used for 3D modelling: Rhino & Fushion 360 & Minecraft
5. Coding using Python for charts.
6. Pictures are pocessed by Photoshop.

Group Work management via Notion

Notion is used for group discussion and data assembling, creating a clear overview of the ideas.

Tasks

Assign tasks ahead to ensure timely work completion and clear role assignment. The 'My Tasks' view only shows the tasks that are assigned to you (the viewer).

All tasks	My tasks
survey prepare and data collection	survey prepare and data collection
observation and report	observation and report
desk research	desk research
ideation for desk research	ideation for desk research

Ideation updated

- material: Wood Polymer Composite, Plastic and recycled material (sustainability)
- ergonomic
- wallrose bench: Flowers on the back of the bench
- solar panel wireless charging
- charging form kinetic energy generating in bikes
- rotating 360
- QR code for history and culture

Interview Summary

Insights

Students

- Aesthetic: (User Attitude)
Users want a more aesthetically pleasing environment because it relaxes and refresh them from studying and working, but they feel overwhelmed that the grass is poorly cared and boring without flowers.
- Insufficient Seats: (User Anxieties)
Insufficient seatings forces students to stand, sit on the floor or search for alternative spaces, which made them less inclined to stay on campus, as their learning and engagement opportunities are limited.
- Not enough place for food after 7pm (User Anxieties)
Students may want more service for customers, students may prefer to work in the library for a long time, but after 19:00, there will be no place for them to get some snack or drinks to replenish energy and keep on working.
- Students would like to have more space for sports on campus because there is a lack of space in Ethos Gym and sometimes the weather is good for sports. They can relax and get rid of the stress for a short time. But they feel concerned about the potential damage that may cause to the grass and environmenting, soil and grass, due to sports.

Bdobe Outdesign

- Meeting Report
- Initial research
- Survey&interview
- observation
- interview
- Tasks
- Meetings
- Calendar
- Insights&Personas
- Teamspace Home
- Ideation
- Ideation updated

Participant Consent Form

Each participant is asked to sign a consent form, showing their consent for participating in the interview.

Participants Consent Form

Full Title of Project: [urban Design for Queen's Lawn]

Please initial box

1. I consent to take part in the above study conducted by a team of students as part of their academic activities at Imperial College.	
2. I confirm that I had the opportunity to ask questions which have been answered fully.	
3. I understand that I am free to withdraw at any time, without giving any reason and without my legal rights being affected.	
4. I understand that the interview will be recorded using a handheld [audio/video] device. I give permission for this recording to take place. I understand that this recording will be deleted once it has been transcribed.	
5. I understand that photographs will be taken and will be anonymized before being included in academic reports.	
6. I understand that anonymised quotations from the interview will be used in academic reports.	

Name of Participant	Signature	Date
Name of Person taking consent	Signature	Date